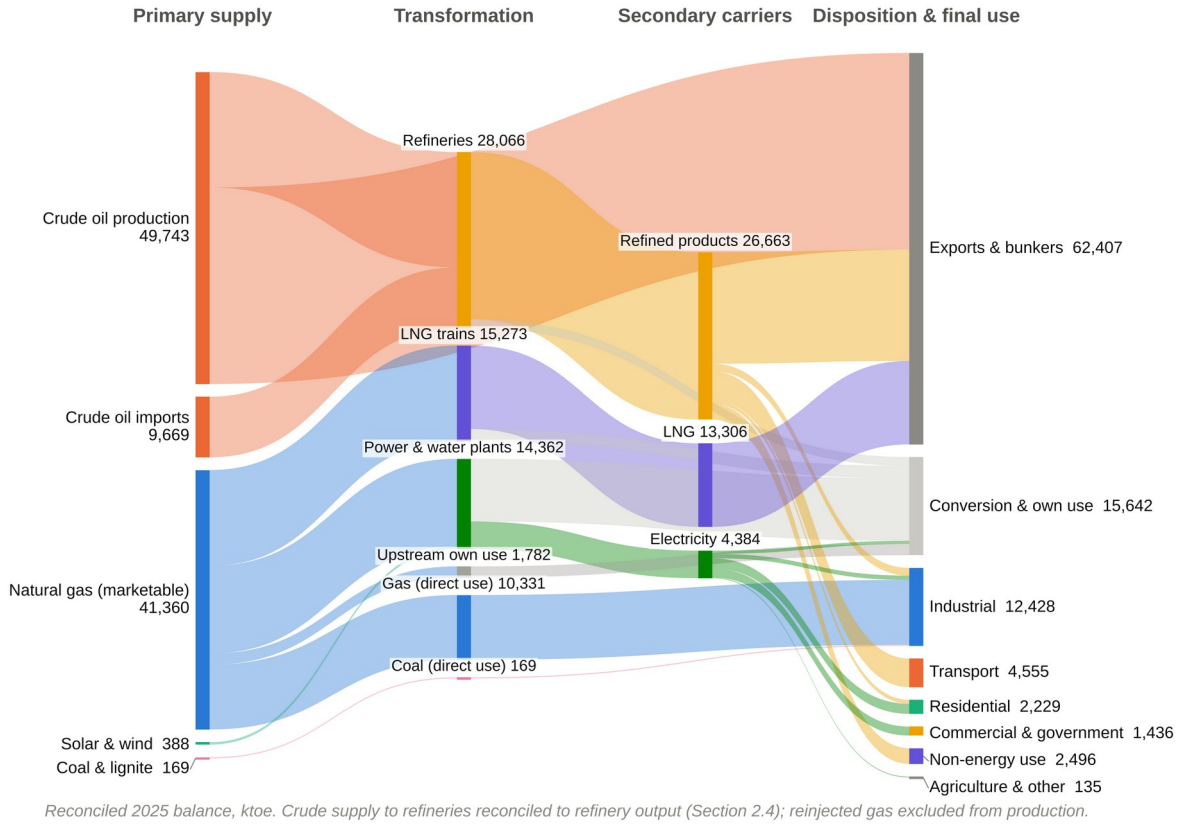
Closing the books of a rentier energy economy: reconciled energy accounts, CO_2 drivers and the renewable dual dividend in Oman, 2015–2025

Highlights

- Closure diagnostics expose refinery, reinjection and power-plant gaps in Oman's data.
- Reconciled 2025 TPES is 38.9 Mtoe, within 1.3% of independent estimates.
- Combustion CO_2 is 88.1 Mt (90% CI 83.4–92.6); gas supplies 74%.
- Population and energy intensity drove CO_2 up 14.1 Mt over 2015–2024.
- Domestic gas use transfers an implicit rent of ~4.7% of GDP a year.

Graphical abstract



Reconciled 2025 balance, ktoe. Crude supply to refineries reconciled to refinery output (Section 2.4); reinjected gas excluded from production.

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Abstract

Hydrocarbon exporters face a distinctive decarbonization problem: the fuels displaced at home are also the source of fiscal rents abroad, and the energy statistics needed to manage this trade-off are often internally inconsistent. This study develops a diagnostic reconciliation framework for producer-state energy balances and applies it to Oman. Column closure, process-efficiency tests and triangulation against independent datasets identify four structural problems in the 2025 national accounts: refinery output exceeding recorded crude intake (apparent efficiency 1.52), reinjected gas counted as production, thermal electricity counted as primary energy, and a large unallocated petroleum residual. After reconciliation, indigenous production is 91,492 ktoe, Total Primary Energy Supply (TPES) 38,923 ktoe (90% CI 37,101–40,669) and the self-sufficiency ratio 235%. TPES lies within 1.3% of independent estimates. Fuel-combustion CO_2 emissions are 88.1 Mt (83.4–92.6), 74% from natural gas. A logarithmic mean Divisia index decomposition of 2015–2024 shows that population growth (+16.9 Mt) and rising energy intensity (+11.6 Mt) outweighed falling carbon intensity (-6.3 Mt). Renewables supplied 8.85% of electricity in 2025, avoiding 1.76–2.39 $MtCO_2$ and freeing 0.96 bcm of gas. Valued at LNG netback, domestic gas combustion transfers an implicit rent of about 4.7% of GDP, of which renewables preserved USD 0.16 billion. Transparent energy accounts are thus a precondition for credible decarbonization in rentier states.

Keywords: Energy balance reconciliation; Rentier state; LMDI decomposition; Monte Carlo uncertainty; Natural gas rents; Oman

Abbreviations

Abbreviation	Definition
bcm	Billion cubic metres
CI	Confidence interval (5th–95th percentile)
EF	Emission factor
EI	Energy Institute
GCC	Gulf Cooperation Council
GCP	Global Carbon Project
IPCC	Intergovernmental Panel on Climate Change
IRES	International Recommendations for Energy Statistics
kbbl	Thousand barrels
ktoe	Thousand tonnes of oil equivalent (1 toe = 41.868 GJ)
LMDI	Logarithmic mean Divisia index

Abbreviation	Definition
LNG	Liquefied natural gas
MMBtu	Million British thermal units
MMscf	Million standard cubic feet
NCV	Net calorific value
NDC	Nationally Determined Contribution
OPWP	Oman Power and Water Procurement Company
SSR	Self-sufficiency ratio (production ÷ TPES)
TFC	Total Final Consumption
TFEE	Total-factor energy efficiency
TPES	Total Primary Energy Supply
VRE	Variable renewable energy

1. Introduction

1.1. Background

The economies of the Gulf Cooperation Council (GCC) are organised around the capture and redistribution of hydrocarbon rents. In the rentier-state model, external rents accrue directly to the state, which allocates them domestically through public employment, transfers and administered energy prices rather than through taxation (Beblawi and Luciani, 1987; Krane, 2019). The same fiscal architecture that sustains political stability also locks in high domestic energy demand: subsidised electricity and fuels have produced per-capita consumption and energy intensities among the highest in the world (Lahn and Stevens, 2011; Fattouh and El-Katiri, 2013; Al-Badi and AlMubarak, 2019). Climate change adds a further structural pressure in a region already exposed to extreme heat and water stress (Al-Maamary et al., 2017a).

The Sultanate of Oman occupies a specific position within this group. It is a mid-sized producer with modest reserves and a history of fiscal deficits, so the opportunity cost of domestic hydrocarbon use is particularly salient (Al-Sarihi and Cherni, 2023). Oman Vision 2040 formalises a pivot towards economic diversification and a low-carbon energy sector. The government has pledged net-zero emissions by 2050 and has updated its Nationally Determined Contribution (NDC) (Sultanate of Oman, 2020, 2021; Argus Media, 2023). The Oman Power and Water Procurement Company (OPWP) targets a 30% renewable share of generation by 2030, and the state-owned Hydrogen Oman (Hydrom) is mandated to develop 1.0–1.25 Mt of renewable hydrogen per year by 2030 (IEA, 2023b; OPWP, 2024).

1.2. Problem statement

Heavily subsidised, centralised petroleum economies exhibit considerable structural inertia. Incumbent institutions, administered prices and long-term export contracts tend to delay the diffusion of renewables even where resources are exceptional (Al-Sarihi and Cherni, 2023; Szalai et al., 2025). A less frequently discussed barrier is informational. Energy and climate policy rests on a national energy balance that traces energy from production and trade through transformation to final use. The balance supplies activity data for emissions inventories, denominators for efficiency indicators and the evidence base for subsidy and investment decisions (UNSD, 2018; Nagovnak et al., 2022).

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Energy balances of hydrocarbon exporters are unusually difficult to compile. Large transformation sectors (refining, liquefaction and gas-fired power-and-water plants) dominate the accounts, so small errors in conversion factors or flow definitions propagate into large errors in primary supply. Flows into special economic zones may be recorded under different trade conventions (UN, 2011). Gas reinjected for pressure maintenance must be separated from gas that is burnt, and administrative data are fragmented across ministries and companies (Neelis and Pouwelse, 2008; Kuunifaa et al., 2026). A balance whose columns sum correctly can still violate physical constraints, such as a refinery that delivers more energy than it receives. Arithmetic closure is therefore a necessary but not a sufficient test of quality.

1.3. Literature review

Four strands of literature frame this study. The first concerns the relationship between growth, energy use and emissions. The Environmental Kuznets Curve (EKC) hypothesis posits an inverted-U relationship between income and environmental pressure (Grossman and Krueger, 1995), but its empirical robustness has been widely questioned (Stern, 2004). For the GCC, panel and time-series studies generally find that energy consumption and emissions remain tightly coupled to growth (Al-Iriani, 2006; Salahuddin and Gow, 2014; Bekhet et al., 2017; Abdel-Gadir and Mohammed, 2024). Index decomposition analysis, in particular the logarithmic mean Divisia index (LMDI), is the standard tool for attributing emission changes to scale and intensity effects (Ang, 2015), yet it has rarely been applied to Oman.

The second strand addresses energy security and the political economy of transition in producer states. Self-sufficiency captures availability but not affordability, sustainability or governance (Sovacool and Mukherjee, 2011; Ang et al., 2015). For exporters, the transition both threatens rents and creates opportunities to monetise low-cost renewable electricity and hydrogen (Goldthau and Westphal, 2019; Gielen et al., 2019). Studies of Oman and the United Arab Emirates show that renewables have been rationalised partly as a means of preserving hydrocarbons for export, attracting investment and projecting a modernised image (Al-Maamary et al., 2017b; Szalai et al., 2025). Using strategic niche management, Al-Sarihi and Cherni (2023) argue that the Omani state supports renewable projects selectively, in ways that preserve the hydrocarbon regime. Regional reviews document GCC targets and readiness gaps (Mondal et al., 2016; Praveen et al., 2020;

Almasri and Narayan, 2021), and Oman-specific work catalogues resource potential, barriers and the hydrogen agenda (Al-Badi et al., 2011; Amoatey et al., 2022; Marzouk, 2024; Leal-Arcas et al., 2025).

The third strand concerns the integration of variable renewable energy (VRE) into thermal grids. As VRE penetration rises, its market value declines and flexibility requirements grow (Denholm and Hand, 2011; Hirth, 2013). In Oman, multi-criteria assessments favour pumped hydro and compressed air storage (Al-Abri et al., 2024), and distributed photovoltaics face financial, technical and awareness barriers (Mishrif and Khan, 2024; Venkatachalam et al., 2025). Transition theory locates such barriers in regimes of institutions, infrastructure and practice rather than in technology alone (Geels, 2002; Sovacool, 2016).

The fourth strand, energy statistics methodology, is the most directly relevant. Nagovnak et al. (2022) show that IRES-based balances must be adapted to capture transition-era flows. Neelis and Pouwelse (2008) document how feedstock and combustion uses are confused in industrial statistics. Liu et al. (2026) demonstrate that territorial Tier 1 accounting misattributes emissions associated with traded fuels, and studies of emerging economies document fragmented activity data and weak quality assurance in national inventories (Kuunifaa et al., 2026). Table 1 summarises the studies most relevant to the present contribution.

Table 1. Synthesis of the literature most relevant to this study.

Study	Scope	Approach	Relevance to this study
Nagovnak et al. (2022)	Austria	IRES balance methodology for the energy transition	Template for matrix structure and flow diagrams
Neelis and Pouwelse (2008)	Netherlands	Reconciling feedstock and combustion in industrial statistics	Precedent for separating LNG feedstock and non-energy use
Ang (2015)	Methodological	LMDI decomposition guide	Decomposition of CO_2 drivers, 2015–2024
Al-Sarihi and Cherni (2023)	Oman	Political economy of renewables in a rentier state	Interpretive frame for rent incidence
Szalai et al. (2025)	UAE, Oman	Political economy of renewable deployment	Renewables as rent preservation
Salahuddin and Gow (2014); Bekhet et al. (2017)	GCC	Econometric growth–energy–emissions nexus	Evidence of persistent coupling
Charabi et al. (2018)	Oman	GHG reduction pathways for a hydrocarbon economy	Policy context for NDC tracking
IEA (2023b); Marzouk (2024)	Oman	Renewable hydrogen and planned capacity mix	Scale of hydrogen-driven VRE build-out

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Study	Scope	Approach	Relevance to this study
Liu et al. (2026)	Global	Origin of traded fuels in combustion CO_2 accounting	Limits of Tier 1 territorial accounting

1.4. Research gap and contribution

International agencies already publish energy data for Oman. The IEA and the United Nations Statistics Division compile balances from national questionnaire returns, and the Energy Institute and the Global Carbon Project report aggregate energy and emission series (IEA, 2023a; Energy Institute, 2025; Our World in Data, 2026). These products do not document how national source data are reconciled, and their aggregate format hides the transformation flows that matter most in a gas-exporting economy. Oman-specific studies, in turn, rely on these aggregates; none tests the physical consistency of the underlying accounts, quantifies their uncertainty or links them to the distribution of hydrocarbon rents (Charabi et al., 2018; Abdel-Gadir and Mohammed, 2024).

This paper makes four contributions. First, it proposes a diagnostic reconciliation framework for producer-state energy balances. The framework combines column closure, process-efficiency closure and triangulation against independent datasets, and it is transferable to other exporters. Second, it applies the framework to Oman's 2025 national data, documents each correction and propagates the remaining uncertainty through a Monte Carlo analysis. Third, it places 2025 in a 2015–2024 context and decomposes the change in CO_2 emissions with the LMDI method. Fourth, it quantifies the political economy of the transition by estimating the implicit gas rent transferred to domestic users and the rent preserved by renewables, the “dual dividend”. Section 2 describes the data and methods, Section 3 presents the results, Section 4 discusses the political economy of the transition, Section 5 sets out policy implications and limitations, and Section 6 concludes.

2. Data and methods

Fig. 1 summarises the research design. We compile activity data, harmonise units and flow definitions, apply closure diagnostics, and use the reconciled accounts for emissions, decomposition, uncertainty and rent analysis. A failed diagnostic triggers re-harmonisation of the source data before analysis rather than being absorbed into a residual.

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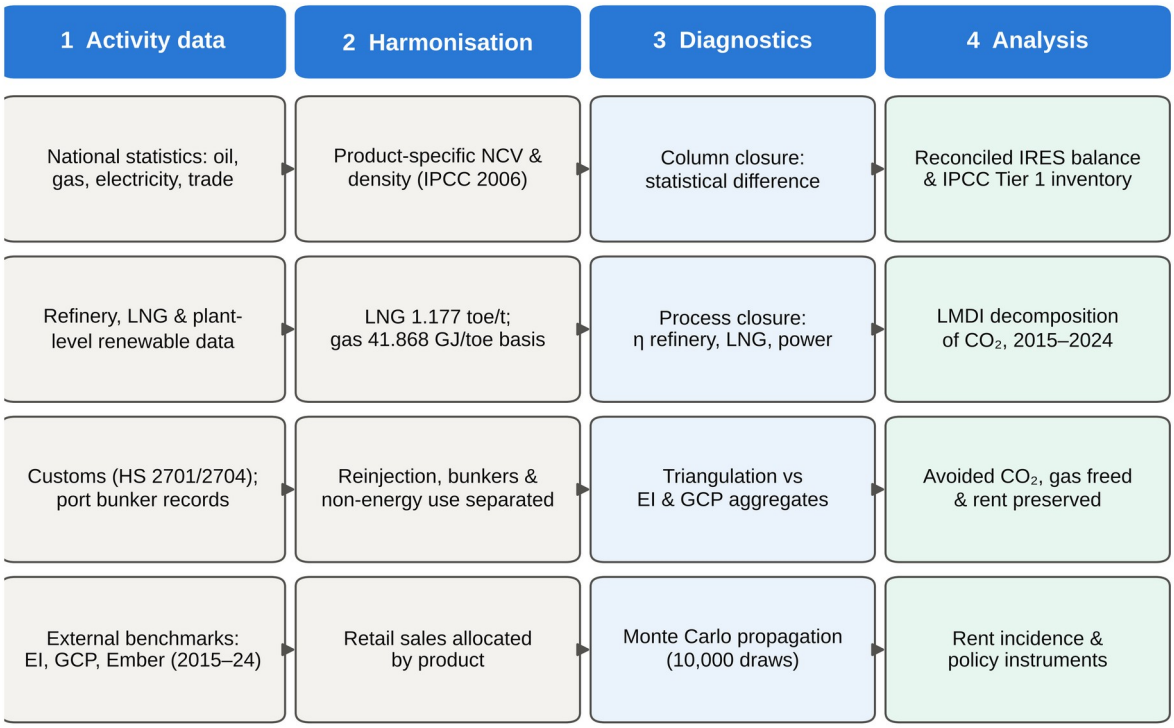


Fig. 1. Research framework: activity data, harmonisation, closure diagnostics and analysis.

2.1. Accounting framework and closure tests

We follow the International Recommendations for Energy Statistics (UNSD, 2018), which are mirrored in IEA and Eurostat practice (IEA, 2023a; Eurostat, 2021). For each commodity *c*, Total Primary Energy Supply is:

$$TPES_c = P_c + M_c - X_c - B_c \pm \Delta S_c \quad (1)$$

where *P* is indigenous production, *M* imports, *X* exports, *B* international marine and aviation bunkers and Δ*S* the stock change. Final consumption follows from the commodity identity:

$$TFC_c = TPES_c - TI_c + TO_c - OU_c - L_c - SD_c \quad (2)$$

where *TI* and *TO* are transformation inputs and outputs, *OU* energy-sector own use, *L* distribution losses and *SD* the statistical difference. We apply three tests. Test T1 requires $|SD_c|$ to be small relative to $TPES_c$. Test T2 requires the apparent efficiency of each transformation process *k* to be physically plausible:

$$\eta_k = \Sigma_c\, TO_{c,k} / \Sigma_c\, TI_{c,k}\,, \quad 0 < \eta_k < \eta_k^{max} \quad (3)$$

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Test T3 compares reconciled aggregates with independent estimates: the Energy Institute's primary energy and gas consumption series and the Global Carbon Project's fuel-specific CO_2 series (Energy Institute, 2025; Our World in Data, 2026). Following IRES, primary electricity comprises only solar and wind generation, and reinjected, flared and vented gas is excluded from natural gas production (UNSD, 2018).

2.2. Data

Table 2 lists the data sources. All 2025 activity data are official national statistics or company data submitted to the national statistical system. The 2015–2024 time series are taken from public international datasets.

Table 2. Data sources.

Source	Data	Coverage
National Centre for Statistics and Information (NCSI, 2026)	Crude oil production and exports; refinery output; gas production and use by category; LNG exports; electricity generation, consumption and sectoral use; GDP; population	2021–2025
Ministry of Energy and Minerals	Monthly crude and condensate production; gas use by sector; refinery output by product	2025
Renewable Energy Bulletin	Monthly renewable generation by plant	2022–2025
Duqm refinery operator	Foreign crude imports; product exports	2025
Customs external trade statistics	Coal (HS 2701) and lignite (HS 2704) trade	2025
Port bunkering records	Marine bunker deliveries at five ports	2024–2025
Energy Institute; Ember; Global Carbon Project (via Our World in Data)	Primary energy, gas and oil consumption, electricity generation and renewables share, CO_2 by fuel, population, GDP	2015–2024
IPCC (2006)	Default NCVs and CO_2 emission factors	International default

2.3. Conversion factors

Crude oil is converted at 0.136 ktoe/kbbl and natural gas at 0.02549 ktoe/MMscf (1 toe = 41.868 GJ). Refined products are converted with product-specific factors derived from IPCC (2006) net calorific values and typical densities (Appendix B). The resulting mix-weighted factor for the refinery slate is 0.1265 ktoe/kbbl, lower than the crude factor, as expected for a light-product slate. LNG is converted at 1.178 toe/t, corresponding to a net calorific value of about 49.3 GJ/t. Electricity is recorded at its physical energy content (0.086 ktoe/GWh) and coal at 0.65 ktoe/kt (Table 3).

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Table 3. Conversion factors.

Commodity	Unit	Factor	Basis
Crude oil	kbbl	0.136 ktoe/kbbl	National crude assay
Refined products (slate average)	kbbl	0.1265 ktoe/kbbl	Product-specific IPCC NCV × density (Appendix B)
Natural gas	MMscf	0.02549 ktoe/MMscf	Standard heat content
LNG	t	1.178 toe/t	NCV ≈ 49.3 GJ/t
Coal and lignite	t	0.00065 ktoe/t	IPCC-consistent default
Electricity	GWh	0.086 ktoe/GWh	Physical energy content

2.4. Reconciliation procedures

2.4.1. Refinery closure

Recorded crude intake to refineries, defined as production less exports plus imports, is 17,529 ktoe. Refinery output is 26,663 ktoe (210,833 kbbl), so the apparent efficiency is 1.52, which fails test T2. Three explanations are possible: over-recorded output (H0), over-recorded crude exports (H1) or under-recorded crude production (H2). H0 can be rejected. If output were constrained to the recorded intake at a plausible efficiency, product supply after recorded product exports and bunkers would be negative, which contradicts observed retail sales of 5,700 ktoe. The discrepancy therefore lies on the crude supply side. We estimate refinery intake from output, using a net refinery energy efficiency η_{ref} of 0.95 (range 0.92–0.98), consistent with refinery fuel and loss rates of 4–8% of throughput (Szklo and Schaeffer, 2007). The implied intake is 28,066 ktoe, which leaves a supply gap of 10,537 ktoe. In the central case (H1), we treat the gap as crude delivered to domestic refineries but recorded within the export series. This is consistent with the recording of deliveries into free zones under the special trade system (UN, 2011), since the Duqm refinery lies in a special economic zone. Under H2, the gap is added to production instead. Both hypotheses give the same TPES but different production and self-sufficiency, and both are reported.

2.4.2. Natural gas: reinjection, LNG and power-and-water plants

Gross gas production is 51,459.2 ktoe (2,018,799 MMscf). Of this, 11,880 ktoe is delivered to oil fields, where gas is used both as fuel and for reinjection in miscible and pressure-maintenance schemes; national statistics do not separate the two uses. We estimate the combusted share by triangulating with independent gas consumption and CO_2 series (Section 3.2). These series are

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consistent with 15% combustion (range 0–30%). The reinjected remainder (10,098 ktoe in the central case) is excluded from production, as IRES requires.

LNG feedstock is the operator's reported feed-gas requirement of 599,187 MMscf (15,273.3 ktoe). This corresponds to 53.0 thousand scf per tonne of LNG, typical of liquefaction plants that use 8–12% of feed gas as fuel (Lee et al., 2018). With LNG at 1.178 toe/t, the 11.3 Mt exported equals 13,306 ktoe, which implies a liquefaction efficiency of 87.1%. The plant loss of 1,967 ktoe is treated as combusted fuel (80–100%).

If all the gas delivered to power-and-water plants (13,974 ktoe) were attributed to the 46,464.5 GWh of thermal generation, the fleet efficiency would be only 28.6%. This is implausible for a fleet dominated by combined-cycle plants. Integrated power-and-water plants also supply heat for thermal desalination, and the reported gas may include auxiliary and captive uses. In the absence of a metered split, we attribute to grid electricity the fuel implied by a net fleet efficiency of 45% (range 40–50%), namely 8,880 ktoe. The remaining 5,094 ktoe is reported separately as "desalination heat and unattributed". Its emissions remain in the inventory, but it does not inflate the electricity emission factor.

2.4.3. Petroleum products: bunkers, retail allocation and non-energy use

Domestic product availability after exports (16,578 ktoe) and marine bunkers (453.8 ktoe) is 9,631 ktoe. Retail distribution sales by product (43,699.8 kbbl) are allocated directly. Gasoline goes to road transport; diesel is split between transport (70%), industry (15%) and households (5%); and jet fuel goes to aviation. Of the jet fuel, an estimated 90% (80–95%) is used by international flights and recorded as aviation bunkers (723 ktoe). The remaining 3,930 ktoe is not sold through retail channels. We allocate it in proportion to the non-retail part of the refinery slate: naphtha and other products, LPG and fuel oil. Naphtha and other products are treated as non-energy use (petrochemical feedstock, bitumen and lubricants) with a share of 0.8 (0.6–1.0), and the remainder as industrial fuel. LPG is allocated to households (70%; 50–90%) and industry, and fuel oil to industry. Non-energy use is reported in TFC but excluded from combustion CO_2, in line with IPCC (2006).

2.5. Emissions

We estimate territorial combustion CO_2 with the IPCC (2006) Tier 1 method on an NCV basis (Appendix A):

$$E_{k,f} = A_{k,f} \times EF_f \tag{4}$$

where A is the fuel combusted in IPCC source category k and EF the default factor. The categories are electricity and desalination (1.A.1.a), refining (1.A.1.b), LNG liquefaction and upstream fuel use (1.A.1.c), manufacturing (1.A.2), transport (1.A.3) and residential use (1.A.4). Flaring, venting and process emissions are outside the scope of the inventory and are discussed in Section 5.2. Grid-electricity emissions are attributed to end-use sectors in proportion to electricity consumption. Emissions avoided by renewables are:

$$\Delta E = G_{RE} \times EF_{marg}\,, \quad EF_{marg} = 56.1\ kgCO_2/GJ \times 3.6 / \eta_{marg} \tag{5}$$

where η_{marg} is the efficiency of the displaced gas-fired plant: 0.45 in the central case, between 0.35 (open cycle) and 0.55 (combined cycle) (Siler-Evans et al., 2012). Gas freed by renewables is $G_{RE} \times 3.6/\eta_{marg}$.

2.6. Uncertainty analysis

We propagate parameter uncertainty with a Monte Carlo simulation of 10,000 draws (Table 4) and report central estimates with 90% intervals. Spearman rank correlations between inputs and outputs identify the parameters that dominate uncertainty.

Table 4. Uncertain parameters in the Monte Carlo analysis.

Parameter	Central	Distribution	Basis
Product energy factors (multiplier)	1.00	U(0.97, 1.03)	Density and NCV variation
Refinery net efficiency η_{ref}	0.95	Tri(0.92, 0.95, 0.98)	Szklo and Schaeffer (2007)
Oil-field gas combusted (share)	0.15	U(0, 0.30)	Triangulation (Section 3.2)
LNG and refinery losses combusted	0.90	U(0.80, 1.00)	Plant fuel vs. losses
Grid fleet efficiency η_{pp}	0.45	Tri(0.40, 0.45, 0.50)	CCGT-dominated fleet
Displaced plant efficiency η_{marg}	0.45	Tri(0.35, 0.45, 0.55)	OCGT–CCGT range
Non-energy share of naphtha/other	0.80	U(0.60, 1.00)	Petrochemicals, bitumen, lubricants
LPG used by households	0.70	U(0.50, 0.90)	Cooking demand
Jet fuel in retail (kbbl); international share	6,000; 0.90	U(4,000, 9,000); U(0.80, 0.95)	Airport throughput
LNG price; domestic gas price; liquefaction cost (USD/MMBtu)	10; 3; 1	U(6, 14); U(2, 4); U(0.5, 1.5)	Market range; Gulf News (2013)

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2.7. Time series and decomposition

For 2015–2024 we use primary energy and gas consumption from the Energy Institute, electricity data from Ember, and CO_2 from fuel combustion and flaring from the Global Carbon Project, all compiled by Our World in Data (Energy Institute, 2025; Ember, 2026; Our World in Data, 2026). Real GDP is extended beyond 2022 with official growth rates of 1.3% in 2023 and 1.7% in 2024 (IMF, 2024; Ministry of Finance, 2025). We decompose the change in CO_2 with the additive LMDI-I method (Ang, 2015) applied to the Kaya identity (Kaya and Yokobori, 1997):

$$C = P \times (G/P) \times (E/G) \times (C/E)\,, \quad \Delta C_x = [(C_T - C_0)/(\ln C_T - \ln C_0)] \times \ln(x_T/x_0) \tag{6}$$

where P is population, G real GDP, E primary energy and C CO_2, and x denotes each of the four factors (population, affluence, energy intensity and carbon intensity).

2.8. Rent incidence and the dual dividend

We value each unit of domestically combusted gas at its export opportunity cost: the LNG price multiplied by the liquefaction efficiency, less the liquefaction cost and the regulated domestic price. Industrial users have paid about USD 3/MMBtu since 2015 (Gulf News, 2013):

$$r = P_{LNG} \times \eta_{LNG} - c_{liq} - P_{dom}\,, \quad Rent_s = Q_s \times r \tag{7}$$

where Q_s is the gas burnt by sector s. The rent preserved by renewables is the freed gas multiplied by r. These are gross opportunity costs that assume spare liquefaction capacity or new export outlets; they indicate the scale of implicit transfers, not realisable revenue.

3. Results

3.1. The reconciled 2025 energy balance

Table 5 presents the reconciled balance, and Fig. 2 traces the corresponding flows. Indigenous production is 91,492 ktoe, of which crude oil contributes 54.4%, marketable natural gas 45.2% and solar and wind 0.4%. After imports of 9,841 ktoe, exports of 61,233 ktoe and bunkers of 1,177 ktoe, TPES is 38,923 ktoe. TFC is 23,280 ktoe (60% of TPES). Every column closes; the only residual is a statistical difference of 0.9 ktoe in natural gas, below 0.01% of gas supply.

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Table 5. Reconciled energy balance of Oman, 2025 (ktoe; central case H1).

Flow (ktoe)	Crude oil	Refined products	Natural gas	LNG	Coal & lignite	Solar & wind	Electricity	Total
Indigenous production	49,742.6	–	41,361.2	–	–	387.8	–	91,491.6
Imports	9,668.9	–	–	–	172.1	–	–	9,841.0
Exports	-31,345.5	-16,578.1	–	-13,305.9	-3.3	–	–	-61,232.8
International marine bunkers	–	-453.8	–	–	–	–	–	-453.8
International aviation bunkers	–	-723.4	–	–	–	–	–	-723.4
TPES	**28,066.0**	**-17,755.4**	**41,361.2**	**-13,305.9**	**168.7**	**387.8**	**–**	**38,922.5**
Oil refineries	-28,066.0	26,662.7	–	–	–	–	–	-1,403.3
Power-and-water plants	–	–	-13,974.0	–	–	-387.8	4,383.7	-9,978.1
LNG plants	–	–	-15,273.3	13,305.9	–	–	–	-1,967.4
Upstream own use	–	–	-1,782.0	–	–	–	–	-1,782.0
Distribution losses	–	–	–	–	–	–	-510.9	-510.9
Statistical difference	0.0	0.0	0.9	0.0	0.0	0.0	0.0	0.9
TFC	**–**	**8,907.4**	**10,331.0**	**–**	**168.7**	**–**	**3,872.8**	**23,280.0**
Industrial	–	1,228.8	10,331.0	–	168.7	–	699.7	12,428.3
Transport	–	4,555.0	–	–	–	–	–	4,555.0
Residential	–	627.7	–	–	–	–	1,601.7	2,229.4
Commercial and government	–	–	–	–	–	–	1,436.2	1,436.2
Agriculture & other	–	–	–	–	–	–	135.1	135.1
Non-energy use	–	2,495.9	–	–	–	–	–	2,495.9

Negative transformation entries are inputs and positive entries outputs. Crude oil exports are reconciled to refinery intake (Section 2.4.1); natural gas production excludes reinjection. Totals may not sum exactly owing to rounding.

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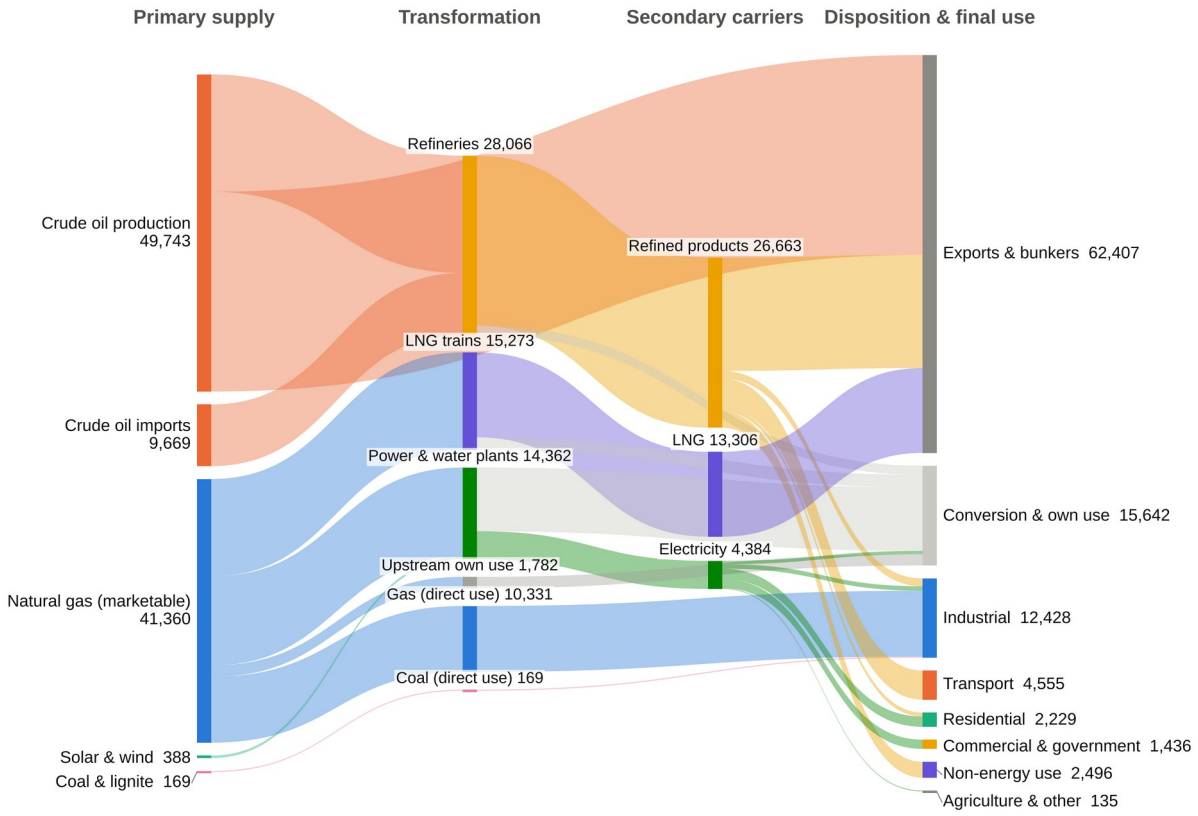


Fig. 2. Energy flows in Oman's reconciled 2025 balance (ktoe).

3.2. Diagnostics and triangulation

Fig. 3 and Table 6 document the diagnostics. Recorded refinery intake falls short of the intake implied by output by 10,537 ktoe (Fig. 3a). The as-compiled TPES of 41,762 ktoe changes through six corrections (Fig. 3b). Restating primary electricity removes 3,996 ktoe, excluding reinjection 10,098 ktoe, the LNG factor 989 ktoe and aviation bunkers 723 ktoe. Product-specific factors lower the energy content of product exports and add 2,430 ktoe, and crude reconciliation adds 10,537 ktoe. The corrections are of similar size and opposite sign, so they largely offset in aggregate (−6.8%), but they substantially change the composition of supply.

Triangulation supports the reconciled accounts (Fig. 3c). Reconciled TPES (38.9 Mtoe) lies within 1.3% of the Energy Institute's 2024 primary energy estimate (38.4 Mtoe). Domestic gas combustion (28.1 Mtoe) lies within 4% of the Energy Institute's gas consumption (27.2 Mtoe), and gas-combustion CO_2 (65.4 Mt) lies within 3% of the Global Carbon Project estimate (63.6 Mt). These matches hold only if most oil-field gas is reinjected; if all 11,880 ktoe were burnt, gas CO_2 would exceed the independent estimate by about 26 Mt. Petroleum CO_2 (22.0 Mt) remains above the Global Carbon Project series (12.4 Mt). Road transport alone accounts for 13.5 Mt on the basis

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of metered retail sales, which suggests that the international series underestimate Oman's oil combustion.

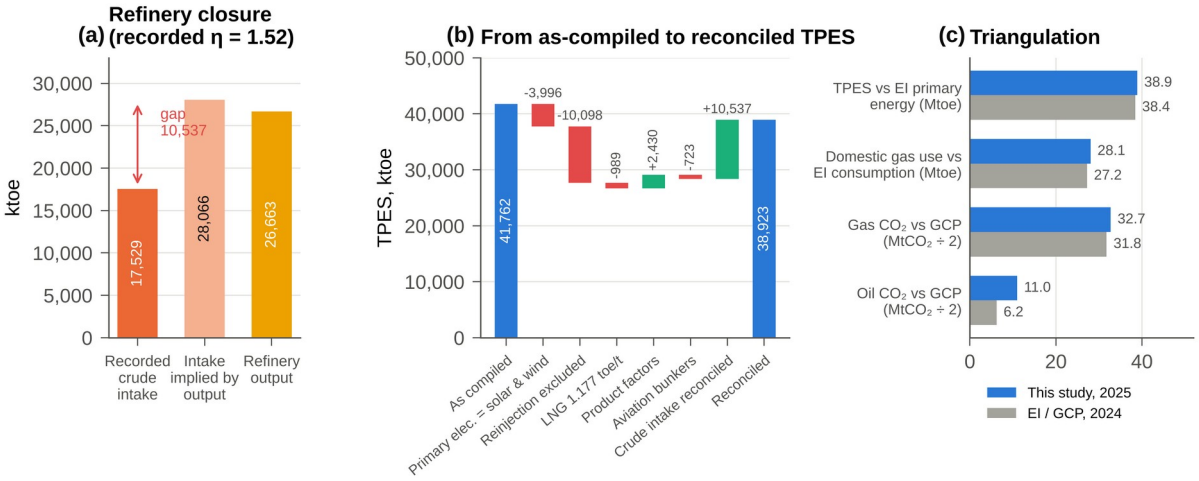


Fig. 3. Closure diagnostics. (a) Refinery closure: recorded crude intake, intake implied by output and refinery output. (b) Reconciliation of TPES from the as-compiled to the reconciled balance. (c) Triangulation against Energy Institute (EI) and Global Carbon Project (GCP) estimates for 2024.

Table 6. Diagnostic findings and treatments.

Issue	Test and finding	Treatment	Effect on TPES (ktoe)
Thermal electricity recorded as primary energy	IRES definition	Primary electricity = solar and wind	−3,996
Oil-field gas includes reinjection	T3: gas CO_2 and consumption exceed EI/GCP if all burnt	15% combusted; remainder excluded from production	−10,098
LNG factor 1.09 toe/t; η_{LNG} 80.6%	T2: below plant range	1.178 toe/t; η_{LNG} = 87.1%	−989
Single product factor 0.145 ktoe/kbbl	Exceeds crude factor	Product-specific factors	+2,430
Jet fuel counted as domestic transport	IRES bunker convention	International share to aviation bunkers	−723
Refinery η = 1.52	T2: physically impossible; H0 rejected	Crude intake reconciled to output (H1/H2)	+10,537
Power-plant fleet η = 28.6%	T2: below CCGT fleet range	Desalination and unattributed fuel separated	0 (allocation only)
Unallocated petroleum residual (21.7% of TFC)	Product composition	Product-based allocation incl. non-energy use	0 (allocation only)

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3.3. Supply: oil, gas and coal

Table 7 compares headline indicators before and after reconciliation. The SSR is 235% under H1 (90% CI 228–243%) and 262% under H2, against 252.8% in the as-compiled data. Under every hypothesis Oman produces more than twice its primary energy requirement.

Table 7. Headline indicators, 2025.

Indicator	As compiled	Reconciled, H1 (90% CI)	Reconciled, H2
Indigenous production (ktoe)	105,585.5	91,492 (89,884–93,096)	102,029
TPES (ktoe)	41,761.7	38,923 (37,101–40,669)	38,923
TFC (ktoe)	25,481.7	23,280 (22,850–23,630)	23,280
Self-sufficiency ratio (%)	252.8	235.1 (228.3–242.9)	262.1
Energy intensity (toe per RO million)	1,040.7	969.9 (925–1,013)	969.9
Per-capita TPES (kgoe)	7,792.0	7,262	7,262
Renewable share of TPES (%)	0.93	1.00	1.00
Fuel-combustion CO_2 (Mt)	91.48	88.1 (83.4–92.6)	88.1

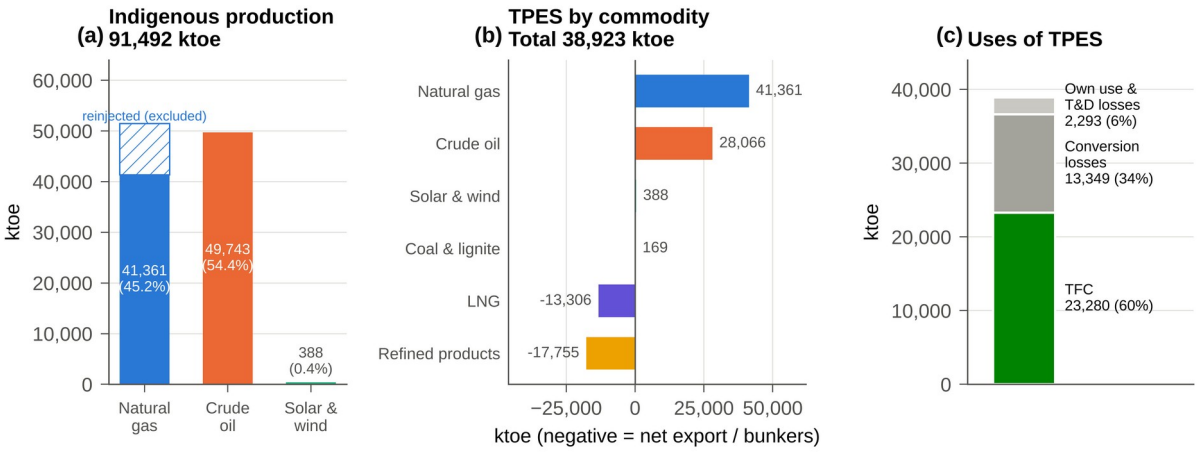


Fig. 4. Supply, 2025. (a) Indigenous production, with reinjected gas excluded. (b) TPES by commodity; negative values denote net exports and bunkers. (c) Uses of TPES.

Crude production averaged 1,002 kbbl/d (Fig. 5a). Recorded exports of 41,883 ktoe fall to 31,345 ktoe after reconciliation. The product-resolved slate has an energy content of 14,561 ktoe and is dominated by diesel (33%) and naphtha and other products (23%) (Fig. 5b). Of gross gas production, 29.7% is liquefied, 27.2% is delivered to power-and-water plants, 19.6% is reinjected, 3.5% is burnt in upstream operations and 20.1% is used by industry (Fig. 6a). Liquefaction operates at 87.1% efficiency (Fig. 6b). Coal and lignite imports, used mainly by cement plants, contribute 168.7 ktoe.

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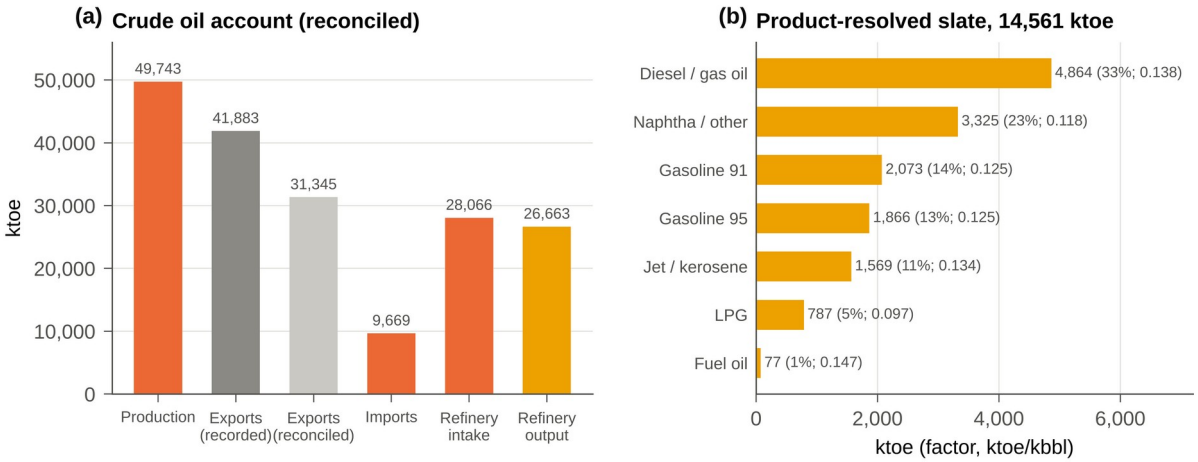


Fig. 5. Oil account, 2025. (a) Crude oil supply and refinery throughput, recorded and reconciled (ktoe). (b) Product-resolved refinery output by energy content, with product-specific factors (ktoe/kbbl).

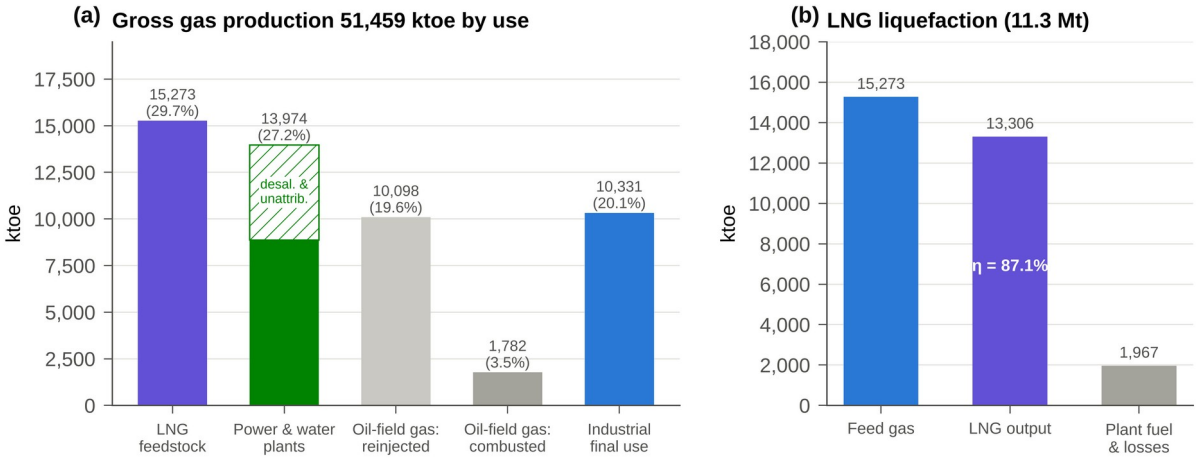


Fig. 6. Natural gas, 2025. (a) Gross production by use; the hatched segment is power-and-water plant fuel not attributable to grid electricity. (b) LNG liquefaction balance.

3.4. Final consumption

Table 8 and Fig. 7 present TFC by sector and carrier. Industry is the largest final consumer (12,428 ktoe; 53.4%), and natural gas meets 83% of its demand. Transport accounts for 19.6% and non-energy use for 10.7%, while residential, commercial and government buildings together account for 15.7%. The product-based allocation replaces the previous unallocated residual of 21.7% of TFC. Its largest remaining uncertainty is the split between non-energy and energy use of naphtha (90% CI for non-energy use: 1,927–3,054 ktoe). Because refinery output and product exports are both observed, TFC is not affected by the crude reconciliation; its 90% interval is 22,850–23,630 ktoe.

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Table 8. Total Final Consumption by sector and carrier, 2025 (ktoe).

Sector	Gas	Petroleum	Electricity	Coal	Total	Share (%)
Industrial	10,331	1,229	700	169	12,428	53.4
Transport	–	4,555	–	–	4,555	19.6
Non-energy use	–	2,496	–	–	2,496	10.7
Residential	–	628	1,602	–	2,229	9.6
Commercial	–	–	1,046	–	1,046	4.5
Government	–	–	390	–	390	1.7
Agriculture & other	–	–	135	–	135	0.6
Total	**10,331**	**8,907**	**3,873**	**169**	**23,280**	**100.0**

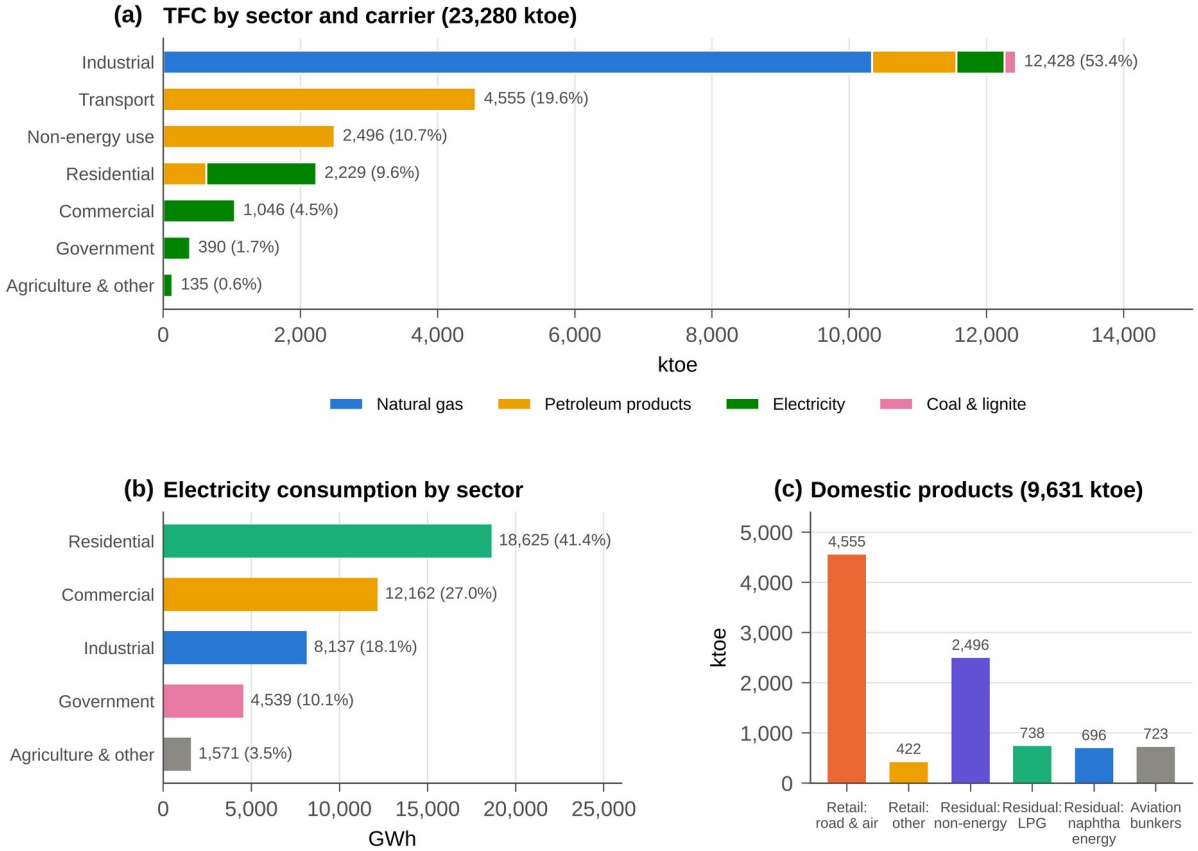


Fig. 7. Final consumption, 2025. (a) TFC by sector and carrier. (b) Electricity consumption by sector. (c) Allocation of domestic petroleum products, including aviation bunkers.

Electricity shows a different structure. Of the 45,033 GWh consumed, households use 41.4%, commercial users 27.0%, industry 18.1% and government 10.1% (Fig. 7b). Renewable generation reached 4,509 GWh, or 8.85% of generation, up from 4.89% in 2024 (Fig. 8). Three photovoltaic plants supplied 97.2% of this output. Preliminary international data for 2025 report only 4.5% because they carry forward the 2024 solar output; the plant-level data used here capture the

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commissioning of new capacity. Reaching the OPWP target of 30% by 2030 requires a gain of 4.2 percentage points a year, whereas the national 20% target assessed by the IEA (2023b) requires 2.2 points a year.

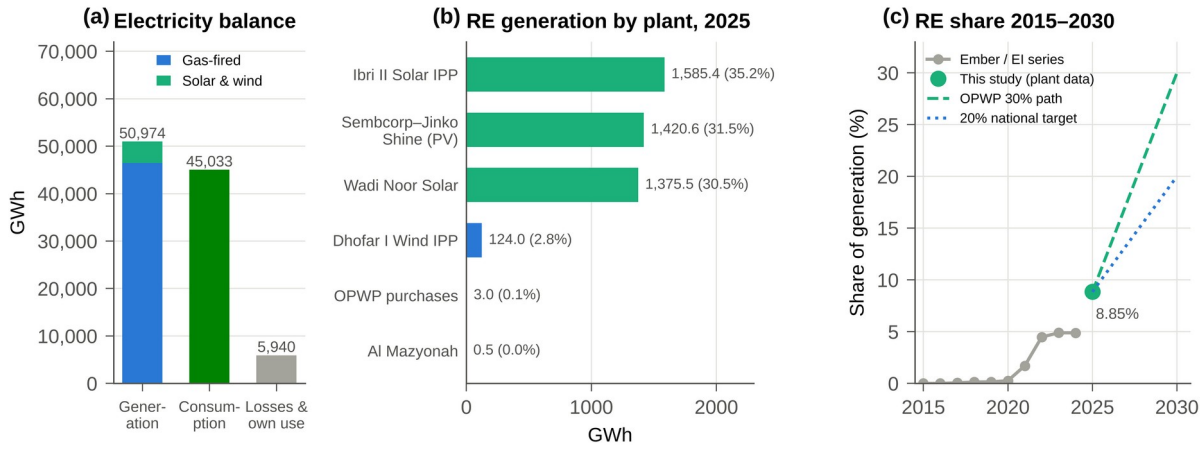


Fig. 8. Electricity, 2025. (a) Electricity balance. (b) Renewable generation by plant. (c) Renewable share of generation, 2015–2025, with the 2030 targets.

3.5. CO_2 emissions

Fuel-combustion CO_2 emissions are 88.1 Mt (90% CI 83.4–92.6), or 16.4 t per capita (Table 9; Fig. 9). Energy industries (1.A.1) account for 44.2 Mt. Of this total, 20.9 Mt arises from grid electricity, 12.0 Mt from desalination and unattributed plant fuel, and 11.4 Mt from refining, liquefaction and upstream fuel use, three categories absent from the as-compiled inventory. Manufacturing contributes 28.7 Mt and transport 13.5 Mt. Natural gas accounts for 74% of emissions, and international bunkers add a memo item of 3.6 Mt. Under end-use attribution, buildings account for 18.1 Mt and the energy sector's own operations for 23.4 Mt.

Table 9. CO_2 emissions from fuel combustion by IPCC category, 2025, compared with the as-compiled inventory (Mt).

IPCC category	Reconciled	Share (%)	As compiled
1.A.1.a Electricity (grid)	20.86	23.7	32.82*
1.A.1.a Desalination heat & unattributed	11.97	13.6	*
1.A.1.b Petroleum refining	3.05	3.5	–
1.A.1.c LNG liquefaction	4.16	4.7	–
1.A.1.c Oil & gas extraction	4.19	4.8	–
1.A.2 Manufacturing industries	28.66	32.5	27.81
1.A.3 Transport	13.52	15.3	13.29
1.A.4 Residential	1.71	1.9	0.95
Unallocated petroleum residual	–	–	16.60

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IPCC category	Reconciled	Share (%)	As compiled
Total	**88.10**	**100.0**	**91.48**
Memo: international bunkers	3.64		–

*The as-compiled inventory attributed all power-and-water plant gas to electricity.

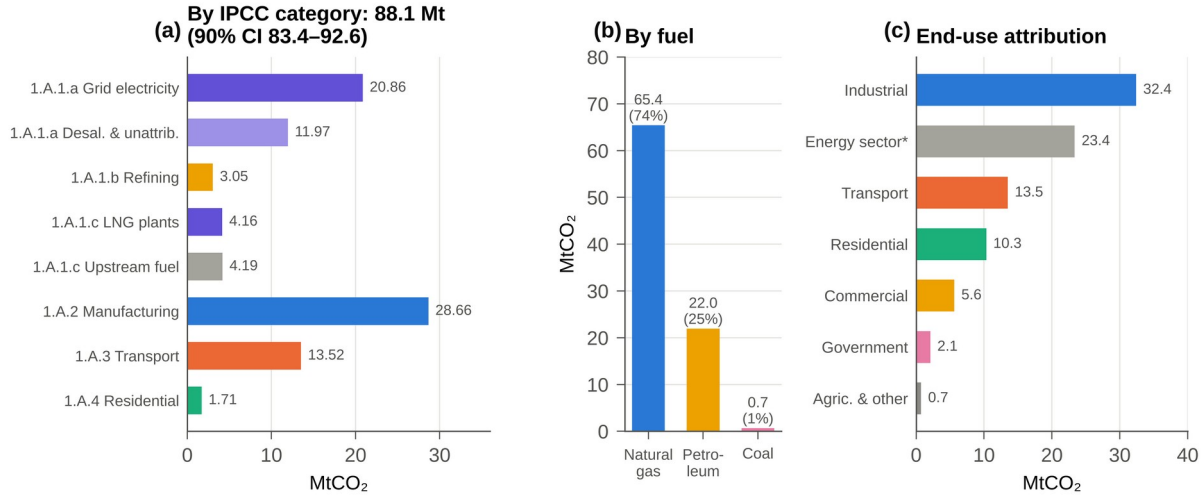


Fig. 9. CO_2 emissions from fuel combustion, 2025. (a) By IPCC category. (b) By fuel. (c) End-use attribution, with grid-electricity emissions allocated by consumption. *Energy sector: upstream, LNG, refining, desalination and unattributed fuel.

3.6. Emission trends and drivers, 2015–2024

CO_2 from fuel combustion and flaring rose from 65.7 Mt in 2015 to 79.8 Mt in 2024 (+14.1 Mt), and the share of gas increased from 73% to 80% (Fig. 10a). The LMDI decomposition (Table 10; Fig. 10b) attributes the increase mainly to population growth (+16.9 Mt) and rising energy intensity of GDP (+11.6 Mt). Falling real GDP per capita (-8.1 Mt) and declining carbon intensity of energy (-6.3 Mt) partly offset these effects. The energy-intensity effect accelerated after 2019 (+8.3 Mt in 2019–2024, against +3.2 Mt in 2015–2019), coinciding with the expansion of gas-intensive industry and the commissioning of the Duqm refinery. Carbon intensity fell mainly through oil-to-gas substitution and, from 2021, through solar generation. These results reject absolute decoupling: emissions grew while real GDP per capita fell, the opposite of the trajectory implied by the EKC (Stern, 2004).

Table 10. LMDI decomposition of the change in CO_2 emissions from fuel combustion and flaring (Mt).

Period	Population	GDP per capita	Energy intensity	Carbon intensity	Total
2015-2019	6.28	-2.64	3.23	-2.80	4.06
2019-2024	10.46	-5.38	8.33	-3.39	10.03
2015-2024	16.89	-8.05	11.55	-6.29	14.09

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The reconciled 2025 inventory (88.1 Mt, excluding flaring) lies above the 2024 series value (76.0 Mt excluding flaring). The difference reflects scope and data rather than a one-year jump: it arises from the higher petroleum emissions implied by national retail data and from upstream, liquefaction and refinery fuel that the aggregate international series may not fully capture.

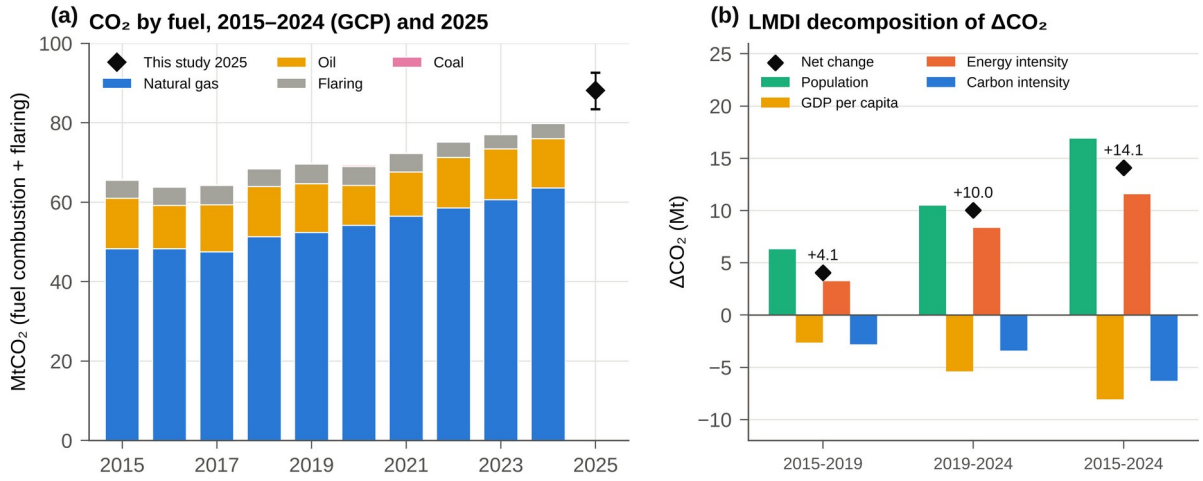


Fig. 10. Emission trends. (a) CO_2 from fuel combustion and flaring by fuel, 2015–2024 (Global Carbon Project), and the reconciled 2025 inventory with its 90% interval. (b) LMDI decomposition of the change in CO_2.

3.7. Uncertainty

Fig. 11 summarises the Monte Carlo results. The 90% intervals are 37,101–40,669 ktoe for TPES (±4.6%), 228–243% for the SSR and 83.4–92.6 Mt for CO_2. The combusted share of oil-field gas dominates the uncertainty in both the SSR (ρ = -0.81) and CO_2 (ρ = +0.85), followed by refinery efficiency and the non-energy share of naphtha. Metered data on reinjection and refinery intake would therefore reduce the remaining uncertainty more than any other improvement.

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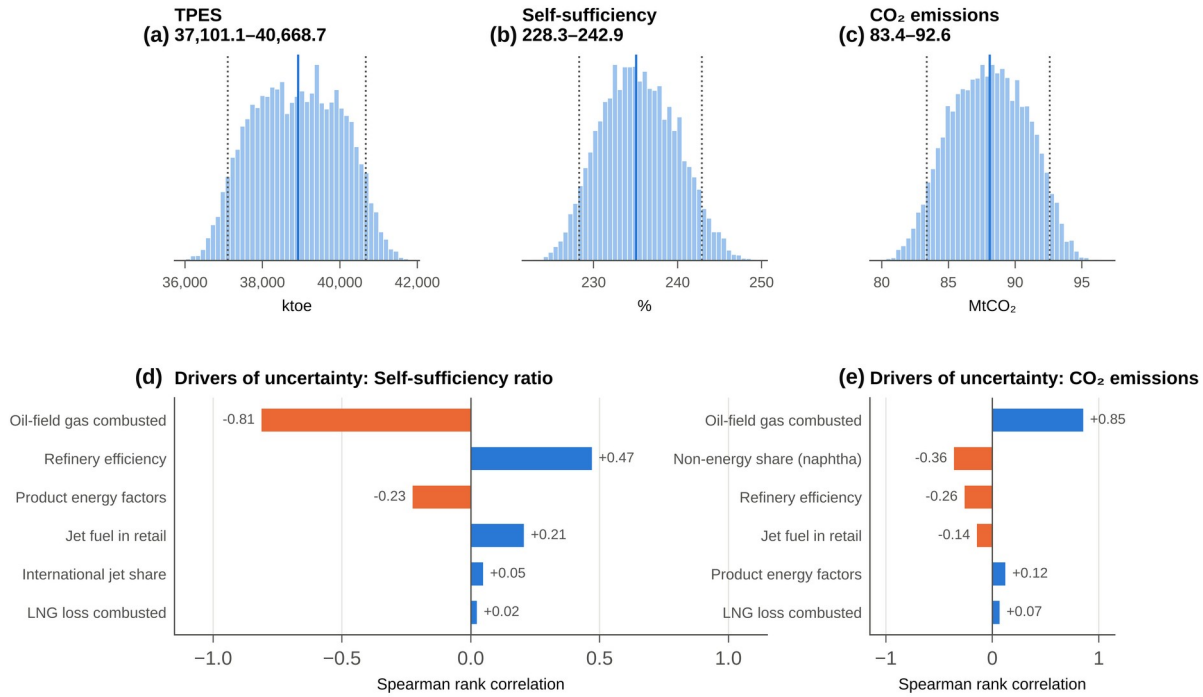


Fig. 11. Monte Carlo uncertainty (10,000 draws). (a–c) Distributions of TPES, the self-sufficiency ratio and CO_2 emissions; solid lines mark the central estimate and dotted lines the 90% interval. (d–e) Spearman rank correlations between parameters and outputs.

4. Discussion: the political economy of Oman's transition

4.1. Rent incidence in a gas-based rentier economy

Rentier theory predicts that the state distributes resource rents to sustain political allegiance. In energy-exporting economies much of this distribution occurs off-budget, through energy sold below its opportunity cost (Beblawi and Luciani, 1987; Krane, 2019). The reconciled accounts allow this transfer to be measured. At an LNG price of USD 10/MMBtu, net of liquefaction losses and costs, domestic gas combustion in 2025 carried an implicit rent of USD 4.9 billion, or 4.7% of GDP (90% CI 1.4–8.0%). Power-and-water plants received USD 2.6 billion, industry 1.9 billion and upstream operations 0.3 billion (Fig. 12a). By comparison, the explicit electricity subsidy in the 2025 budget is RO 520 million, or USD 1.35 billion (Gulf News, 2025). The implicit gas rent is thus several times larger than the visible subsidy, and most of it reaches consumers through electricity tariffs and industrial gas prices. This helps explain why tariff and gas-price reform is politically sensitive. It is also consistent with the argument of Al-Sarihi and Cherni (2023) that renewable deployment has been embraced selectively, in forms that leave these distributional channels intact.

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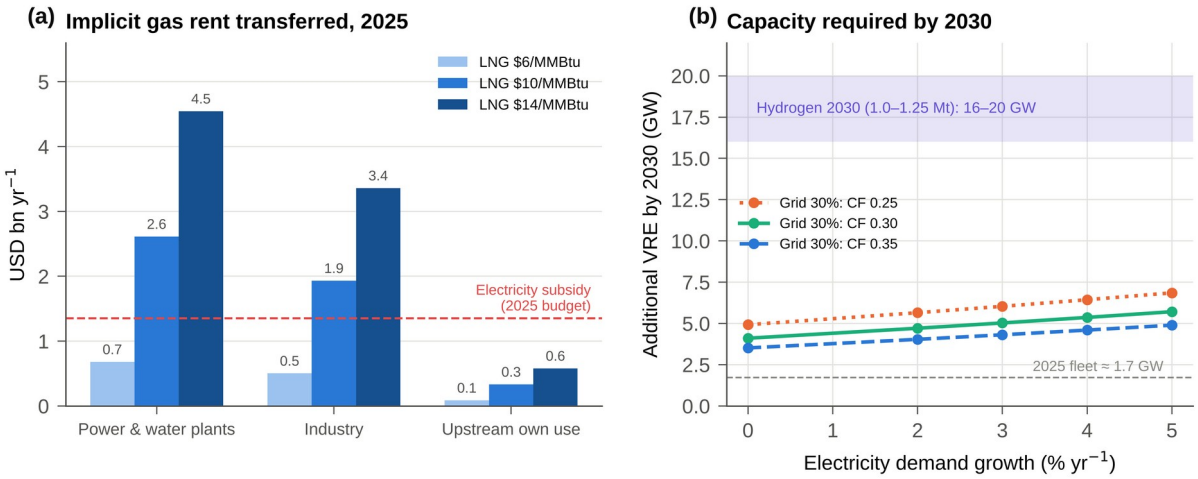


Fig. 12. Political economy and scale. (a) Implicit gas rent transferred to domestic users in 2025 at three LNG prices, compared with the explicit electricity subsidy. (b) Additional VRE capacity required for a 30% renewable share by 2030, by demand growth and capacity factor, compared with the capacity required for the 2030 hydrogen target.

4.2. The dual dividend of renewables

In 2025 renewables avoided 2.02 $MtCO_2$ (90% CI 1.76–2.39) and freed 0.96 bcm of gas (Fig. 13a). The 862 ktoe of freed gas preserved an export rent of USD 0.16 billion (0.05–0.28). With 30% renewables and constant generation, avoided emissions would reach 5.6–8.8 Mt and the preserved rent USD 0.14–0.95 billion per year, depending on prices (Fig. 13b). This dual dividend of abatement and rent preservation is specific to exporters; for importers the main benefit of renewables is a lower import bill (Al-Maamary et al., 2017b; Szalai et al., 2025). The dividend aligns climate policy with the fiscal interests of the rentier state, which helps explain why renewables have gained support among incumbents. It has two limitations, however. Gas that is exported or redirected to new domestic industry is still burnt elsewhere, so global emissions fall by less than the territorial saving (Davis et al., 2011; Liu et al., 2026). And the preserved rent can be realised only if liquefaction capacity or other export outlets are available.

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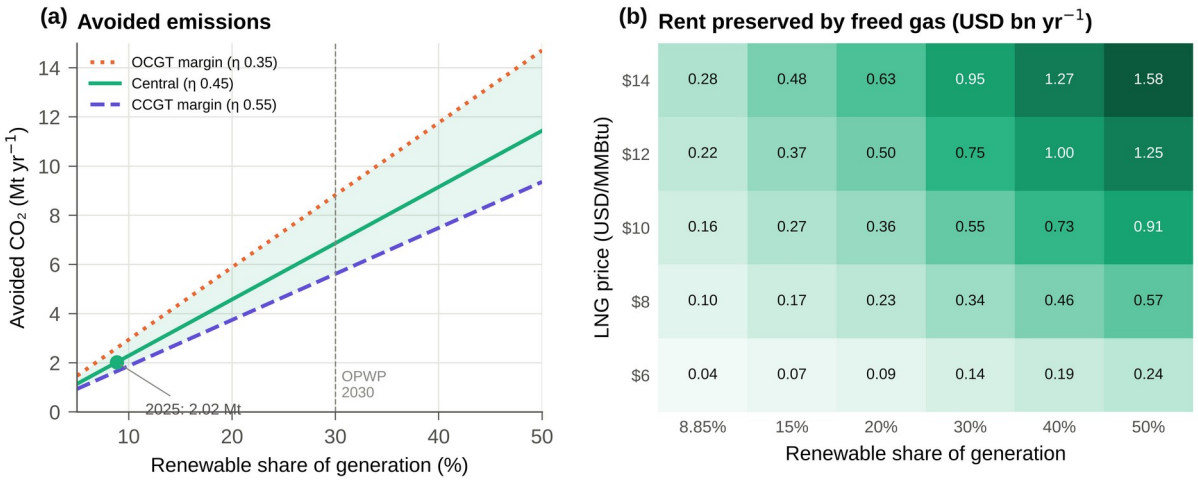


Fig. 13. The dual dividend. (a) Avoided CO_2 as a function of renewable share for the central and bounding displaced-plant efficiencies, with 2025 generation held constant. (b) Export rent preserved by freed gas as a function of renewable share and LNG price (domestic price USD 3/MMBtu; liquefaction cost USD 1/MMBtu).

4.3. Demand-side management and energy efficiency

The decomposition shows that rising energy intensity added 11.6 Mt of CO_2 between 2015 and 2024, more than the abatement that renewables could deliver at the 30% target. Energy efficiency has historically been under-prioritised in the GCC because administered prices weaken the incentive to conserve (Fattouh and El-Katiri, 2013; Almasri and Narayan, 2021). Partial indicators such as energy per unit of GDP conflate structural change with efficiency (Patterson, 1996). The total-factor energy efficiency (TFEE) framework, which treats energy as one input alongside labour and capital, provides a better benchmark for the industrial gas users identified here (Hu and Wang, 2006; Tanaka, 2008). On the electricity side, buildings consume 78.5% of electricity, largely for cooling. Cost-reflective tariffs with targeted household support, building codes and monitoring of heating, ventilation and air-conditioning are the instruments most directly supported by the evidence. Each additional percentage point of annual demand growth raises the VRE capacity needed for the 30% target by about 0.3 GW (Fig. 12b).

4.4. Hydrogen and the scale of the build-out

Hydrom's targets of 1.0–1.25 Mt of hydrogen by 2030, 3.75 Mt by 2040 and 8.5 Mt by 2050 imply a build-out of a different order of magnitude from the grid transition (IEA, 2023b). Assuming about 55 kWh per kg of hydrogen and a hybrid solar–wind capacity factor of 0.40, the 2030 target requires 55–69 TWh of electricity, or 16–20 GW of dedicated capacity. This compares with an implied 2025 VRE fleet of about 1.7 GW and grid additions of 3.5–7 GW for the 30%

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target (Fig. 12b). Marzouk (2024) shows that planned projects would raise renewables to about 83% of a total of 83 GW, almost entirely because of hydrogen. Wind-to-hydrogen studies confirm that resources are favourable (Ahshan et al., 2022), and the IEA estimates cumulative investment of about USD 33 billion for 1 Mt by 2030 (IEA, 2023b). The reconciled balance supplies the accounting framework needed to verify additionality, that is, to confirm that electricity for electrolysis does not displace grid supply and increase gas burn elsewhere (Amoatey et al., 2025).

4.5. Green finance and value-chain localisation

The capital required exceeds what the state budget can supply under volatile oil prices. Mobilising private capital depends on credible taxonomies, disclosure and bankable offtake. Omani green bonds, sukuk and sustainable finance frameworks indicate growing market depth, although overlapping agency mandates raise transaction costs (Ali, 2025). A green taxonomy aligned with established classifications such as the EU taxonomy (European Union, 2020) would reduce greenwashing risk. Value-chain localisation is a complementary hedge: a planned USD 565 million photovoltaic plant in the Sohar Freezone, with 6 GW of cell and 3 GW of module capacity, would place Oman upstream in the solar value chain (pv magazine, 2024; Enterprise AM, 2025). Lithium brines at Umm as Samim and Mahout could anchor a transition-minerals sector (Mishrif and Khan, 2023; IEA, 2021). For such strategies to support decarbonization rather than simply add energy-intensive industry, their energy demand must be tracked in the balance from the outset.

4.6. Implications for energy statistics in producer states

The diagnostics generalise beyond Oman. Producer-state balances are dominated by transformation flows in which small definitional errors become large errors in primary supply. In Oman's case, six corrections ranging from 0.7 to 10.5 Mtoe largely offset one another, leaving a net change of 7%, so agreement with international aggregates does not by itself establish the quality of the underlying accounts. Process-efficiency tests (T2) and triangulation of fuel-specific emissions (T3) detect errors that column closure (T1) cannot. We recommend that producer states report refinery intake, reinjection and power-and-water fuel splits as metered quantities, and that international compilers request them explicitly (UNSD, 2018; Kuunifaa et al., 2026).

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5. Policy implications and limitations

5.1. Policy recommendations

Fig. 14 organises the recommendations into supply-side, demand-side and institutional pillars resting on a common evidence layer, and Table 11 links each instrument to the evidence that motivates it.

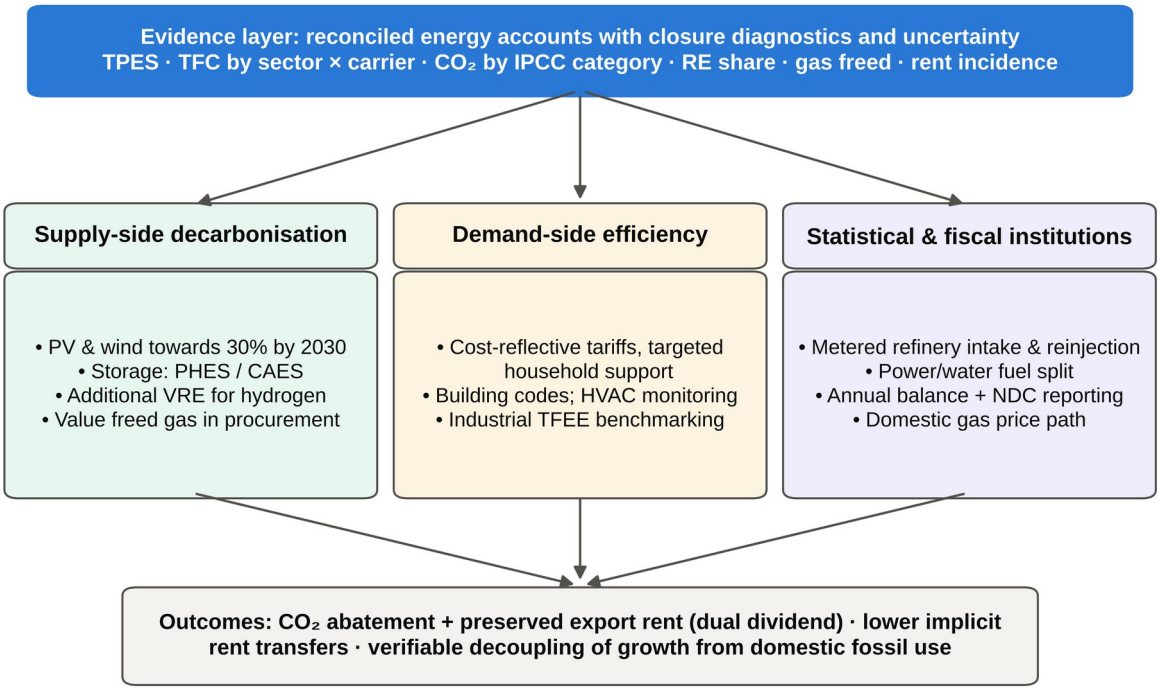


Fig. 14. Policy framework linking the reconciled evidence layer to supply-side, demand-side and institutional instruments and to transition outcomes.

Table 11. Policy instruments and supporting evidence.

Instrument	Evidence	Lead institutions	Horizon
Meter refinery crude intake, reinjection and power-and-water fuel splits	Gaps of 10.5 Mtoe (refinery), 10.1 Mtoe (reinjection) and 5.1 Mtoe (power-and-water)	Ministry of Energy and Minerals; NCSI; operators	Short term
Adopt the reconciled balance as the basis for NDC reporting	Inventory of 88.1 Mt, including 1.A.1 categories previously missing	Environment Authority; NCSI	Short term
Value freed gas in renewable procurement	0.96 bcm freed; USD 0.16 billion preserved in 2025	OPWP; Ministry of Finance	Short term
Gradual domestic gas price path towards opportunity cost	Implicit rent ≈ 4.7% of GDP	Ministry of Finance; regulator	Medium term
Industrial TFEE benchmarking and fuel switching	Industry 53% of TFC; 28.7 Mt CO_2	Ministry of Energy and Minerals; industrial zones	Medium term
Cost-reflective tariffs with targeted support; building codes	Buildings 78.5% of electricity	Regulator; municipalities	Short to medium term

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Instrument	Evidence	Lead institutions	Horizon
Additionality rules for hydrogen	16–20 GW for hydrogen versus 3.5–7 GW for the grid	Hydrom; OPWP	Medium term

5.2. Limitations and future work

Table 12 catalogues the remaining limitations. The refinery reconciliation rests on an assumed net efficiency and on the attribution of the gap to crude exports (H1) or production (H2); metered intake data would replace both assumptions. The combusted share of oil-field gas, the split of power-and-water fuel and the non-energy share of naphtha are estimated rather than measured, and the Monte Carlo analysis quantifies but cannot remove their effect. The Tier 1 inventory excludes flaring (about 3.8 Mt in 2024 according to the Global Carbon Project), venting, fugitive methane and process emissions, all significant in an upstream-intensive economy (Elvidge et al., 2018). The 2015–2024 decomposition relies on international series whose definitions differ from the national balance, so the 2025 point is not strictly comparable with the series. Future work should build a national balance time series on the reconciled basis and move key categories to Tier 2.

Table 12. Remaining limitations and required data.

Limitation	Effect	Required data
Refinery crude intake not metered	Production and SSR depend on H1/H2; TPES varies with η_{ref}	Refinery intake by refinery and crude origin
Reinjection not separated from fuel use	Dominant source of CO_2 and SSR uncertainty	Field-level reinjection and fuel gas
Power-and-water fuel split	5.1 Mtoe of fuel not attributed to grid electricity	Plant-level fuel, power and water output
Non-energy use of naphtha	Non-energy use 1,927–3,054 ktoe	Petrochemical feedstock statistics
Bunkers incomplete	Marine bunkers indicative	Port-level reconciliation
Tier 1 scope	Flaring, venting, fugitive and process CO_2 excluded	Tier 2/3 upstream and cement inventory

6. Conclusions

This paper has developed a diagnostic reconciliation framework for producer-state energy balances and applied it to Oman. Arithmetic closure concealed four structural problems in the 2025 accounts: refinery output exceeding recorded crude intake, reinjected gas counted as production, thermal electricity counted as primary energy, and a large unallocated petroleum residual. After reconciliation, Oman produces 91.5 Mtoe against a primary supply of 38.9 Mtoe, a self-sufficiency

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ratio of 235% (262% if the refinery gap reflects under-recorded production). Fuel-combustion emissions are 88.1 $MtCO_2$, 74% of which derive from natural gas. Between 2015 and 2024 emissions grew by 14.1 Mt, driven by population growth and rising energy intensity, with no evidence of decoupling.

Renewables reached 8.85% of generation in 2025, avoiding about 2 $MtCO_2$ and freeing about 1 bcm of gas. This dual dividend aligns climate policy with the fiscal interests of a rentier state, but it is small relative to the implicit gas rent of about 4.7% of GDP transferred to domestic users. Deep decarbonization will therefore depend less on supply-side expansion alone than on reform of the domestic pricing and statistical institutions through which that rent is distributed and measured. Annually maintained, transparent energy accounts are a precondition for verifying progress towards Oman Vision 2040 and the 2050 net-zero target.

Data availability

The reconciled balance, Monte Carlo outputs, time series and all scripts needed to reproduce the figures and tables are deposited at Zenodo (https://doi.org/10.5281/zenodo.XXXXXXX) under a CC BY 4.0 licence. Confidential company-level source data are not included; aggregate inputs are provided in the data package.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) for drafting support, figure scripting and language editing. The authors reviewed and edited all content and take full responsibility for the publication.

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Appendix A. Emission factors

Table A.1. IPCC (2006) default CO_2 emission factors (NCV basis).

Fuel	Carbon content (kg C/GJ)	EF (kg CO_2/TJ)	EF (t CO_2/toe)
Crude oil	20.0	73,300	3.0689
Motor gasoline	18.9	69,300	2.9015
Jet kerosene	19.5	71,500	2.9936
Gas/diesel oil	20.2	74,100	3.1024
LPG	17.2	63,100	2.6419
Residual fuel oil	21.1	77,400	3.2406
Naphtha/other	20.0	73,300	3.0689
Refinery gas	15.7	57,600	2.4116
Natural gas	15.3	56,100	2.3488
Anthracite/bituminous coal	26.8	98,300	4.1156
Lignite	27.6	101,000	4.2287

Source: IPCC (2006), Volume 2, Chapter 1, Table 1.4.

Appendix B. Product-specific conversion factors

Table B.1. Product-specific conversion factors and the product-resolved refinery slate.

Product	NCV (GJ/t)	Density (t/m³)	Factor (ktoe/kbbl)	Output (kbbl)	Output (ktoe)
Diesel / gas oil	43.0	0.845	0.1380	35,254	4,864
Naphtha / other	44.5	0.700	0.1183	28,111	3,325
Gasoline 91	44.3	0.745	0.1253	16,540	2,073
Gasoline 95	44.3	0.745	0.1253	14,887	1,866
Jet / kerosene	44.1	0.800	0.1340	11,712	1,569
LPG	47.3	0.540	0.0970	8,112	787
Fuel oil	40.4	0.960	0.1473	524	77
Slate (mix-weighted)			0.1265	115,140	14,561

NCV from IPCC (2006); densities are typical values. Factor = density × 0.158987 m³/bbl × NCV / 41.868 GJ/toe.

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